Predictive Classification In HR Data

Documentation & Analysis

Majid M. Ben Ghet

1 Introduction

Our study will use the IBM attrition dataset with the target variable "Attrition" being the binary variable of whether an employee is retained or attrited (stay or leave).

2 Preprocessing and visualization

In the preprocessing stage we have applied many techniques to handle irregularity with the dataset. Firstly, we removed four variables that are constant—like the hours and the fact that they are not minors—which are all the same, additionally dropping the identifier variable. Secondly, we checked for missing values but there were none, hence no action was needed in that regard.

We also applied a few feature engineering techniques to some of the variables. For instance, the income variables are right-skewed. This skewness can negatively impact the linear models we will apply. For this reason, we will have to log-transform them. Other feature engineering we will apply is the codifying of the ordinal variables, like the traveling frequency, to retain the ordinal nature of these variables.

Moreover, we had to handle the outliers of the dataset and determine whether they raised a bias in the data, so we checked the skewness of the variables to determine the method we want to use to impute them.

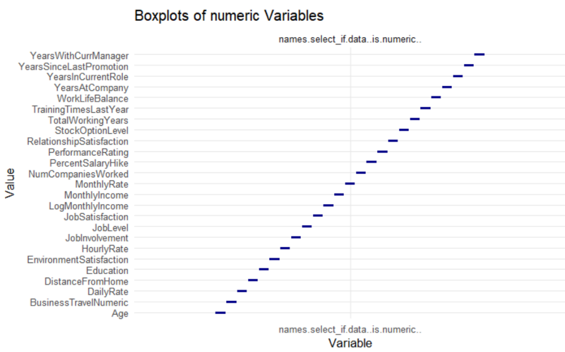

Figure 1: Insert appropriate caption for Boxplots/Correlation here.

From the boxplots above we can see that many of our variables are skewed, and since the 3-Sigma method assumes normality of the data, we cannot use it. We will proceed with using the 3*IQR method to remove the outliers, retaining 93% of the dataset.

We now visualized some of the variables which seemed to have had some importance towards our study:

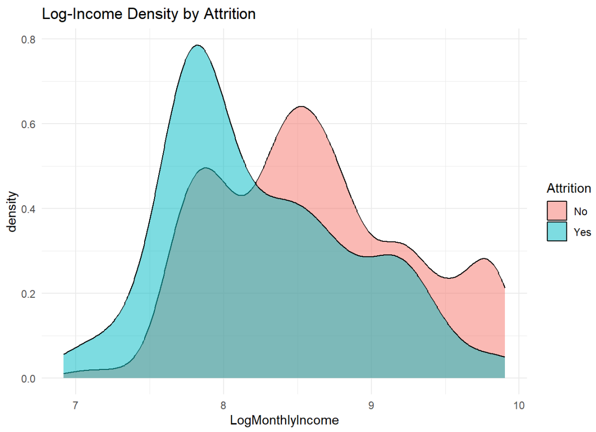

Figure 2: Income and Age Density Distributions

This graph shows that when employees have higher income there is a higher chance of staying. On the other hand, we can also say that lower income results in them leaving the company; there is a turning point where, beyond that point in the income scale, people are more likely to stay. The age with attrition density graph shows the "yes" group peaks at ages 28 to 32, which indicates that people in the early years of their career have more chance of leaving or being let go.

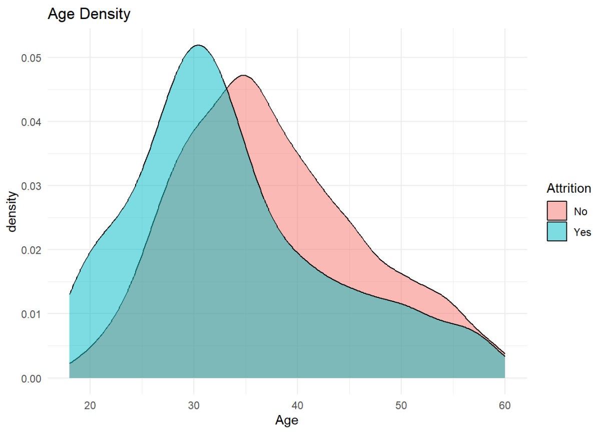

Figure 3: Attrition Rates by OverTime and Job Role

The overtime barplot shows that employees who had done overtime were 3 times more prone to being attrited. The Job role with attrition barplot shows that more stress-inducing roles usually result in higher attrition, in addition to showing that the role of Sales Executive is extremely unstable.

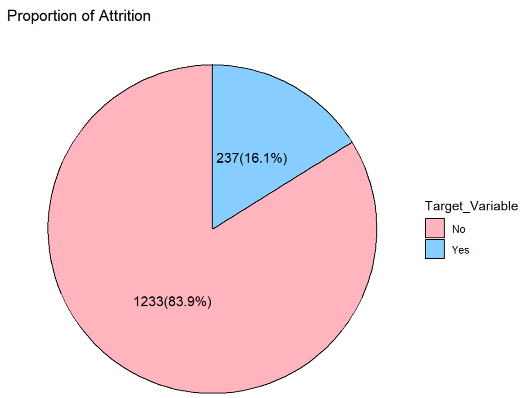

Figure 4: Overall Attrition Proportion

3 Predictive Classification

The biplot analysis indicates a severely skewed target distribution, with approximately 84% of the dataset belonging to the "No" (retained) category. Consequently, models optimizing strictly for overall accuracy will inevitably exhibit bias towards the majority class. For a robust evaluation, it is critical to prioritize metrics such as sensitivity, recall, and the Area Under the Curve (AUC) to accurately capture the minority "Yes" (attrited) class.

Furthermore, the correlation matrix reveals near-perfect positive correlations among variables such as Monthly Income, JobLevel, and TotalWorkingYears, corroborating that higher compensation aligns with tenure and experience. Similarly, high collinearity exists between YearsAtCompany, YearsInCurrentRole, and YearsWithCurrManager. This redundancy introduces high variance in standard regression algorithms, suggesting that ensemble-based models like Random Forests may yield superior stability and performance. It is worth noting that model performance was evaluated both prior to and following feature selection procedures to assess variable significance.

3.1 Probabilistic Models

3.1.1 Logistic Regression

Logistic regression serves as an interpretable baseline model, utilizing odds ratios to determine feature significance. However, it struggles significantly with our highly imbalanced dataset. The model achieved 86% accuracy and 95% sensitivity, but its specificity was a poor 38%, demonstrating a severe inability to correctly identify employees who actually leave. Furthermore, applying feature selection degraded performance, indicating that the discarded features retained critical predictive value. While the AUC is strong at 0.852, the default 0.5 probability cutoff is inappropriately high for this skewed distribution.

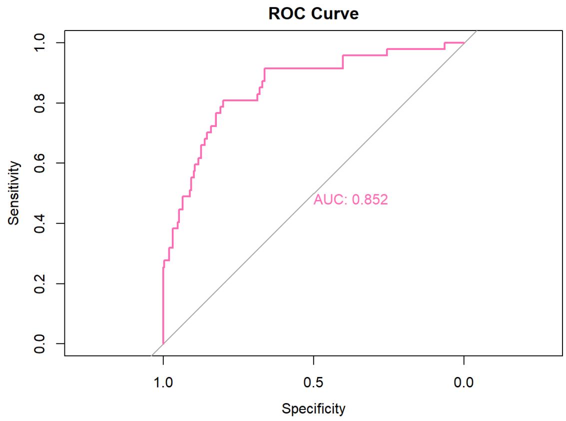

Figure 5: ROC Curve: Logistic Regression

3.1.2 Penalized Logistic Regression

To address the variance and overfitting potential of the baseline model, Penalized Logistic Regression applies regularization constraints. This variant achieved slight improvements: overall accuracy rose to 88.4%, and the Cohen's Kappa score improved from 0.39 (Fair) to 0.46 (Moderate), indicating a more reliable reduction in false positives. Despite a marginally higher AUC, specificity remained tied with the baseline at 38%. The model's reliance on feature selection yielded similarly underwhelming improvements.

3.1.3 Linear Discriminant Analysis (LDA)

LDA focuses on maximizing the separability between classes by projecting the data into a lower-dimensional space. This model outperformed the logistic regression variants on multiple fronts. It achieved a higher specificity of 40.4% while maintaining a competitive 86% accuracy and an AUC of 0.85. Within the probabilistic category, LDA stands as the most balanced and effective predictive model, though it also suffered a slight performance drop when restricted only to the "optimal" selected variables.

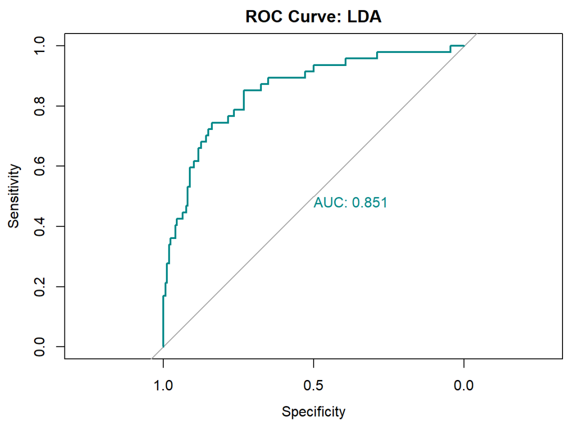

Figure 6: ROC Curve: Linear Discriminant Analysis

3.1.4 Quadratic Discriminant Analysis (QDA)

Unlike LDA, QDA models each class with its own distinct covariance matrix, allowing for more flexible, non-linear decision boundaries. However, QDA relies heavily on the assumption of low multicollinearity among predictors. Given the highly correlated features within our dataset, the QDA algorithm failed to converge effectively on both the full dataset and the reduced feature set.

3.1.5 Naïve Bayes

Operating under the strict assumption of conditional independence among features, Naïve Bayes provided a distinct tradeoff. The model successfully resolved the minority class identification issue, achieving a drastically improved specificity of 72.34%. This success comes at the cost of overall accuracy, which dropped to 79.18%, with an AUC of 0.800. For an organization prioritizing the identification of high-risk flight candidates over overall accuracy, this risk-sensitive model is highly viable.

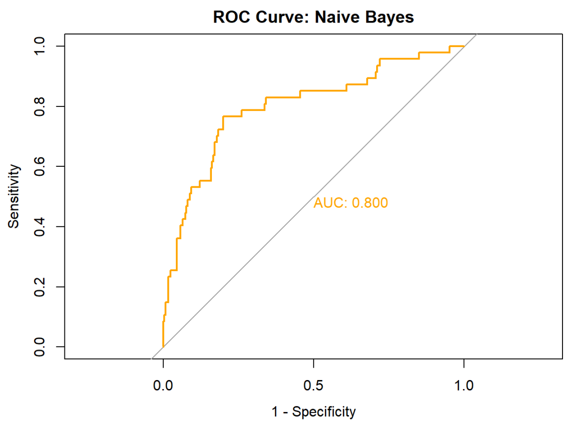

Figure 7: ROC Curve: Naïve Bayes

3.2 Interpretable & Machine Learning Classification

3.2.1 Decision Trees

Decision trees classify data through hierarchical, transparent rules, offering high interpretability for management strategies. The estimated tree identified TotalWorkingYears and OverTime as the most critical primary splits, emphasizing career stage and workload fatigue. Subsequent branches isolated LogMonthlyIncome, Marital Status, and JobRole, highlighting economic and demographic drivers. Predictively, the model yielded a moderate AUC of 0.717. While it suffers from the bias-variance trade-off—sacrificing absolute accuracy for transparency—its structural clarity is invaluable for developing actionable retention protocols.

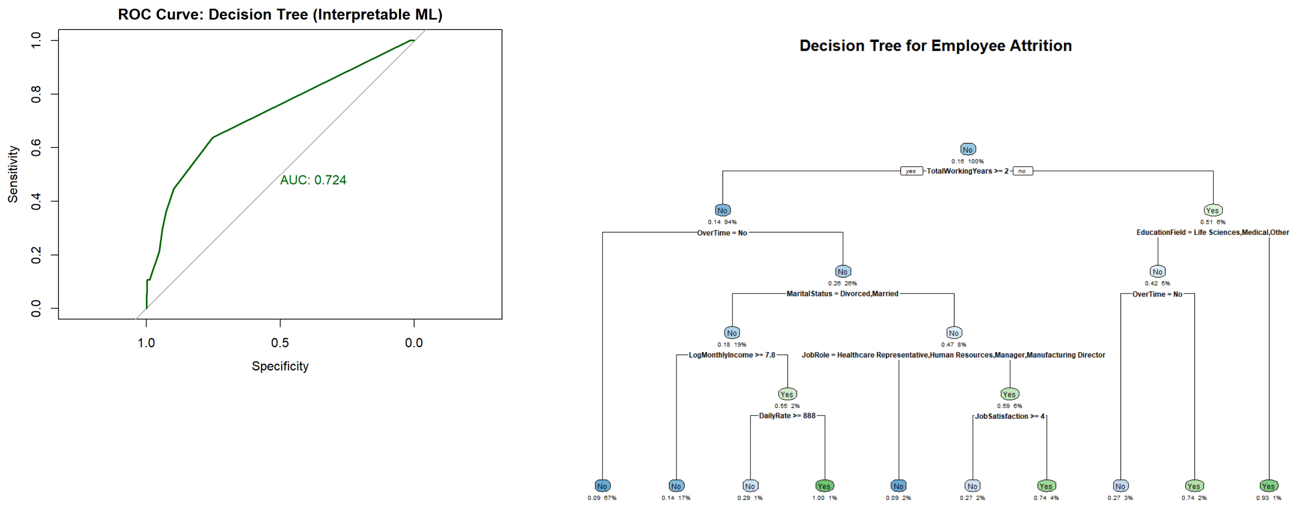

Figure 8: Decision Tree Visualization and ROC Curve

3.2.2 Random Forest

To mitigate the high variance and overfitting issues of a singular decision tree, the Random Forest ensemble averages multiple unpruned trees. This approach secured an impressive AUC of 0.832 and an accuracy of 0.856, positioning it near the top of all evaluated models. However, this predictive superiority inherently sacrifices explicit rule interpretability, rendering it less actionable for direct HR intervention strategies despite its robust statistical validity.

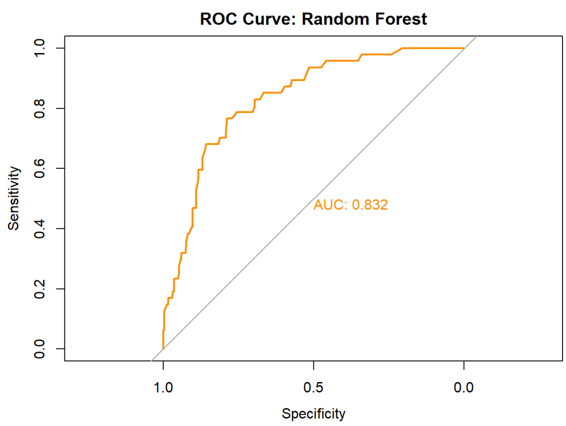

Figure 9: ROC Curve: Random Forest

3.2.3 K-Nearest Neighbors (KNN)

KNN captures complex, non-linear boundaries by referencing local geometric proximity. Optimized at $k = 13$, the model achieved a competitive 85.4% accuracy and an AUC of 0.701 on the full dataset. Yet, because distance metrics are highly sensitive to feature availability, running KNN on the reduced feature space further degraded its AUC to 0.688.

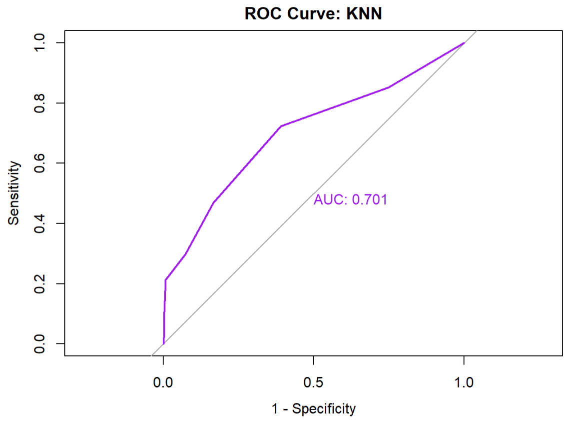

Figure 10: ROC Curve: K-Nearest Neighbors

3.3 New Case Study

To validate the practical utility of our optimal models (Penalized Logistic Regression, LDA, and Random Forest), we synthesized profiles for two theoretical employees: one constructed with high-risk attrition attributes and one with low-risk attributes.

The high-risk inputs included features such as frequent business travel, significant overtime, and lower monthly income relative to their peers. The low-risk profile included non-traveling, standard hours, and higher compensation. Across all three models, the high-risk employee was accurately predicted to attrite with extreme confidence (probabilities ranging from 83% to 99%). Conversely, the low-risk employee received a minimal 5% probability of attrition. This consistent consensus confirms that the isolated variables—OverTime, JobRole, and Income—are statistically robust and reliable indicators across differing algorithmic methodologies.

3.4 Conclusions

Within the probabilistic modeling domain, Penalized Logistic Regression proved superior in balancing accuracy and variance control, closely trailed by LDA. Naïve Bayes, while mathematically constrained, demonstrated unique utility for risk-sensitive applications due to its unparalleled specificity. Notably, aggressive feature selection proved detrimental across the board, establishing that the majority of our variables maintain substantive predictive weight.

Among the non-linear machine learning architectures, Random Forest delivered the highest overall accuracy and stability, adeptly managing the dataset's internal correlations without falling prey to the overfitting seen in singular Decision Trees. Ultimately, Random Forest provides the most rigorous predictive framework for identifying potential attrition, while the baseline logistic and tree models remain indispensable for translating those mathematical predictions into interpretable, human-readable corporate strategies.